

Sửa chữa và thêm các function của layout 3 giao diện quản lý apartment building parking lot

Sử lý bên

Mainactivity.kt

Source code

```
package com.example.appchungcphnngi3
```

```
import android.os.Bundle
import android.text.Editable
import android.text.TextWatcher
import android.view.View
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var adapter: ArrayAdapter<String>
    lateinit var vehicleList: ArrayList<String>
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        // ===== ÁNH XẠ =====
```

```
        val btnShowList = findViewById<Button>(R.id.btnShowList)
        val btnAdd = findViewById<Button>(R.id.btnAddVehicle)
        val layoutList = findViewById<LinearLayout>(R.id.layoutListContainer)
        val listView = findViewById<ListView>(R.id.listVehicle)
        val edtSearch = findViewById<EditText>(R.id.edtSearch)
```

```
// ===== DỮ LIỆU =====
```

```
vehicleList = arrayListOf(  
    "51A-12345 - Nguyễn Văn A",  
    "59B-67890 - Trần Thị B"  
)
```

```
adapter = ArrayAdapter(this, android.R.layout.simple_list_item_1, vehicleList)  
listView.adapter = adapter
```

```
// ===== HIỆN / ẨN LIST =====
```

```
btnShowList.setOnClickListener {  
    if (layoutList.visibility == View.GONE) {  
        layoutList.visibility = View.VISIBLE  
        btnShowList.text = "Ẩn danh sách"  
    } else {  
        layoutList.visibility = View.GONE  
        btnShowList.text = "Hiển thị danh sách xe"  
    }  
}
```

```
// ===== SEARCH =====
```

```
edtSearch.addTextChangedListener(object : TextWatcher {  
    override fun afterTextChanged(s: Editable?) {}  
    override fun beforeTextChanged(s: CharSequence?, start: Int, count: Int, after: Int) {}  
    override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int) {  
        adapter.filter.filter(s) // lọc theo biển số hoặc tên  
    }  
})
```

```
// ===== THÊM XE =====
```

```
btnAdd.setOnClickListener {  
  
    val builder = android.app.AlertDialog.Builder(this)  
    builder.setTitle("Thêm xe")  
  
    val layout = LinearLayout(this)  
    layout.orientation = LinearLayout.VERTICAL  
    layout.setPadding(50, 20, 50, 10)
```

```

val edtPlate = EditText(this)
edtPlate.hint = "Biển số xe"

val edtOwner = EditText(this)
edtOwner.hint = "Tên chủ xe"

layout.addView(edtPlate)
layout.addView(edtOwner)

builder.setView(layout)

builder.setPositiveButton("Thêm") { _, _ ->
    val plate = edtPlate.text.toString().trim()
    val owner = edtOwner.text.toString().trim()

    if (plate.isNotEmpty() && owner.isNotEmpty()) {

        val newVehicle = "$plate - $owner"

        // thêm vào list
        vehicleList.add(newVehicle)

        // cập nhật listview
        adapter.notifyDataSetChanged()

        Toast.makeText(this, "Đã thêm xe!", Toast.LENGTH_SHORT).show()

    } else {
        Toast.makeText(this, "Nhập đầy đủ!", Toast.LENGTH_SHORT).show()
    }
}

builder.setNegativeButton("Hủy", null)
builder.show()
}
}
}

```

activity_main.xml

<RelativeLayout

```
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp">
```

<!-- Tiêu đề -->

<TextView

```
    android:id="@+id/title"
    android:text="Apartment Building Parking Lot"
    android:textSize="20sp"
    android:textStyle="bold"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:background="@drawable/parking_border"
    android:padding="10dp"
    android:layout_centerHorizontal="true"/>
```

<!-- Button thêm xe -->

<Button

```
    android:id="@+id/btnAddVehicle"
    android:text="Thêm xe"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/title"
    android:layout_marginTop="10dp"/>
```

<!-- Button hiển thị list -->

<Button

```
    android:id="@+id/btnShowList"
    android:text="Hiển thị danh sách xe"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
android:layout_below="@id/btnAddVehicle"
android:layout_marginTop="10dp"/>
```

<!-- KHU VỰC LIST -->

```
<LinearLayout
    android:id="@+id/layoutListContainer"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_below="@id/btnShowList"
    android:layout_marginTop="10dp"
    android:visibility="gone">
```

<!-- Ô tìm kiếm -->

```
<EditText
    android:id="@+id/edtSearch"
    android:hint="Nhập biển số xe..."
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="15dp"
    android:background="@android:drawable/edit_text"/>
```

<!-- ListView -->

```
<ListView
    android:id="@+id/listVehicle"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_marginTop="10dp"/>
```

```
</LinearLayout>
```

<!-- Box trạng thái -->

```
<LinearLayout
    android:id="@+id/layoutStatus"
    android:orientation="vertical"
    android:padding="10dp"
    android:background="@android:drawable/dialog_holo_light_frame"
    android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
android:layout_alignParentBottom="true"
android:layout_alignParentEnd="true">
```

```
<TextView
    android:id="@+id/tvAvailable"
    android:text="Còn: 0/100"
    android:textStyle="bold"
    android:textSize="16sp"
    android:textColor="#4CAF50"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
<TextView
    android:id="@+id/tvStatus"
    android:text="Available"
    android:textStyle="bold"
    android:textSize="14sp"
    android:textColor="#4CAF50"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

```
</RelativeLayout>
```